GPU-Accelerated Continuous Subgraph Matching

Why BFS Beats DFS on the SIMT Architecture

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Amazon (AZ) 8v

LiveJournal

Continuous Subgraph Matching (CSM)

$G \quad \Delta = (v_1, v_2, \ell) \quad q$

-
    - & — Bytedance BG3 [9]
    - — Twitter motif detection [1]
    - & [4]
- **CPU** GraphFlow [2], TurboFlux [3], SymBi [7], CaLiG [8], NewSP [6]
- **ParaCOSM (ICPP'25)** [5] 5 CPU
    - Inner-update
    - Inter-updatesafe

→ GPU

DFS BFS

A DFS

CPU

- ✗
- ✗ Hub
- ✗

B BFS

- ✓ Warp lockstep
- ✓ Prefix-scan
- ✓

(algorithm, query) **BFS** **DFS** GPU `gpu_bfs_versioned`

GPU

- `gpu_classifier` mode `-m gpu`
  safe / unsafe
- `gpu_candidate_filter` PFM2
  query candidate
- `gpu_search` mode `-m gpu_all`
  DFS
- `gpu_bfs_search` modes `-m gpu_bfs[_versioned]`
  BFS device memory

Versioned unsafe BFS

Versioned BFS

1. unsafe (v_1, v_2) frontier
2. partial-match `ver_id`
3. query join + filter
4. prefix-scan coalesced partial-match buffer
5. Buffer $\rightarrow$ flush

/ / CSR BFS batch

Amazon (AZ) 8v

- A100-SXM4-80GB $\times$ 1
- Xeon Platinum 8470, 96
- 1.7 TB DRAM

- oneAPI 2025.3 (icpx)
- CUDA 12.9, sm_80

&

- AZ $|V| \approx 0.4$M, $|E| \approx 3.4$M
- LJ $|V| \approx 4.85$M, $|E| \approx 38.6$M
- 100 8-
- 1800 s timeout / pair

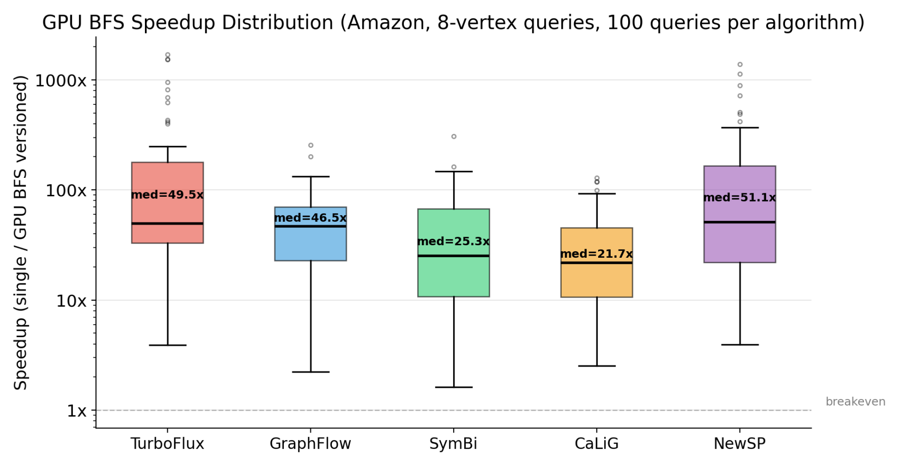
CPU `-m single -t 1`

AZ 8v5

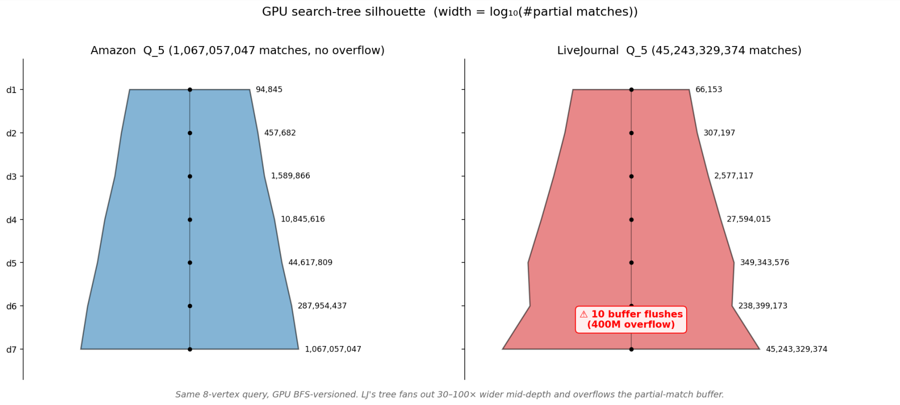
PARALLEL_NEWSP	**51.1**×
PARALLEL_TURBOFLUX	49.5×
PARALLEL_GRAPHFLOW	46.5×
PARALLEL_SYMBI	25.3×
PARALLEL_CALIG	21.7×

CaLiG CPU candidate-set CPU PCIe + kernel-launch

AZ 8v


GPU BFS Speedup Distribution (Amazon, 8-vertex queries, 100 queries per algorithm)
Speedup (single / GPU BFS versioned)
1000x
100x
10x
1x
med=49.5x
med=46.5x
med=25.3x
med=21.7x
med=51.1x
breakeven
TurboFlux
GraphFlow
SymBi
CaLiG
NewSP

BFS — AZ Q_5 (GraphFlow)



(depth 4–5)

partial-match 4×10^8 bufferBFS *flush*

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LJ 8vpicture turns mixed

	pairs	cpu (s)	gpu (s)	
GraphFlow	1	1494.6	105.4	14.18×
TurboFlux	9	666.3	215.2	3.10×
CaLiG	14	583.6	658.9	0.89×
NewSP*	16	87.1	637.0	0.14×
SymBi	0	—	—	—

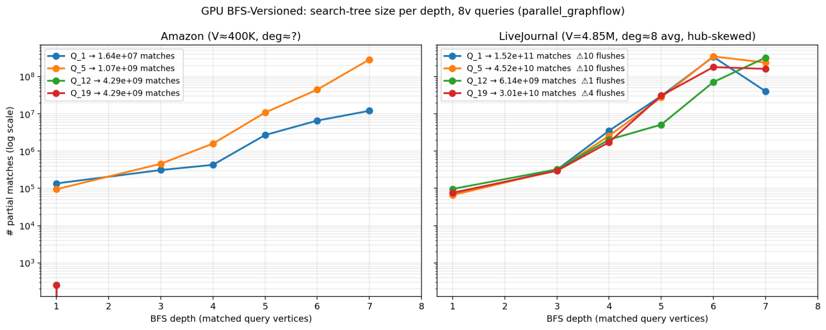
*NewSP CPU `single` 0 adapter stub—

NewSP CPU stub bug

```
size_t Parallel_NewSP_Adapter::EnumerateNewEdge(
        uint v1, uint v2, uint label, size_t /*tid*/) {
  // Not implemented for NewSP adapter -- use versioned mode
  return 0;
}
```

- CPU `single` matching → CPU
- 4 GPU CPU 7 → BFS
- CPU `-m versioned`

LJ



LJ AZ

LJ NewSP `Q_19` 7.97×10^{10} partial-match buffer **flush 15**

log

```
[BFS-V] Depth 5->6: 4.83e7 -> 3.85e8
[BFS-V] Flush 400000000 partials at depth 6->7  (x15)
Search complete: 7.97e10 matches, 1019.0 s
```

device buffer flush BFS stall

Short-term

- **Edge-centric**
  binary-search row-split hub
- **Partial-match streaming**
  buffer D2H pinned memory BFS
  $\rightarrow$ LJ flush thrashing
- **H2D / search overlap**
  stream BFS

Medium-term

- **DFS $\leftrightarrow$ BFS**
  BFS $\rightarrow$ DFS
  query $+$ BFS
- **Early intersection pruning**
  partial-match backward-neighbor
  BFSLJ

Long-term GPU & CPU/GPU

- **GPU**
  - Query GPU query
  - Update safe-update batch GPU
- **CPU / GPU**
  - Safe update → CPUunsafe → GPU
  - NewSP CPU `versioned`
  - data-driven

Take-aways

1. GPU CSM **BFS** DFS
2. AZ 8vBFS-versioned 5 SOTA $21\times$ — $51\times$
3. LJ device buffer H2D flush
4. partial-match profiling
   GPU graph mining workload

ParaCOSM, branch `feat/gpu-profiling-charts`
& `docs/gpu_csm_report/`

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